

SAFE WORK METHOD STATEMENT

Persons responsible for supervising, inspecting work, work methods, plant & equipment and tools	Supervisor/Foreman:	Date:	
High Risk Construction Work:	PLUMBING EXCAVATION & PLUMBING GROUND WORKS	Project:	
SWMS prepared by Patrick Monaghan - Paneltec Group -Operations Manager and in consultation with management and workers – 2025. SWMS Review Date: August 2026 unless required earlier		Location:	
Approval to use this SWMS: "I certify that the attached SWMS has been reviewed by me, is endorsed for compliance with WorkSafe guidelines and is approved for use"			
<u>John Guy</u> Name	<u>Director - Paneltec Pty Ltd</u> Position	<u>0408 449 023</u> 	<u>31/08/2025 – V12.0</u> Date

RISK MATRIX

Level	Description of Consequence or Impact	Consequence	Likelihood / Probability		
			L <i>Likely</i>	M <i>Moderate</i>	U <i>Unlikely</i>
H (1) <i>(High level of harm)</i>	Potential death, permanent disability or major structural failure/damage. Off-site environmental discharge/release not contained and significant long-term environmental harm.	H (1) <i>(High)</i>	1	1	2
M (2) <i>(Medium level of harm)</i>	Potential temporary disability or minor structural failure/damage. On-site environmental discharge/release contained, minor remediation required, short-term environmental harm.	M (2) <i>(Medium)</i>	1	2	3
L (3) <i>(Low level of harm)</i>	Incident that has the potential to cause persons to require first aid. On-site environmental discharge/release immediately contained, minor level clean up with no short-term environmental harm.	L (3) <i>(Low)</i>	2	3	3
Level	Likelihood / Probability				
Likely	Could happen frequently				
Moderate	Could happen occasionally				
Unlikely	May occur only in exceptional circumstances				

HIERARCHY OF RISK CONTROL

The hierarchy of controls must be applied in the order below for all identified hazards.

Level 1 Eliminate the hazard
Level 2 Substitute the hazard with something safer.
Level 3 Isolate the hazard from people. Reduce the risks through engineering controls.
Level 4 Reduce exposure to the hazard using administrative actions.
Level 5 Use personal protective equipment
The control measures that you put in place should be reviewed regularly to make sure they work as planned

ACTIVITY ANALYSIS

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Training and skill.	Un-trained persons could cause injury to themselves and others.	1	1. All persons working on site must carry with them a current 'Construction Induction Card' (Red Card or equivalent). 2. All persons working on site must be suitably trained and competent in the task they are to perform.	Worker	2
Personal Protective Clothing & Equipment (PPE)	Personal injury	1	1. Before starting work on site check available PPE is adequate for all tasks to be under taken and in good working condition including; safety boots, safety glasses, ear muffs and other protective clothing & equipment as required.	Worker	2
UV Protection	UV Exposure; skin cancer.	1	1. Be sun smart wear a broad brimmed hat, sun/safety glasses, sun protective clothing and 30+ sunscreen.	Worker	2
Vehicle access	Vehicle Collisions with other workers or pedestrians.	1	1. Comply with speed limits on and around site. 2. Drive only in designated access routes. 3. Do not park vehicles where it will create a hazard to public vehicles, pedestrians or other workers.	Worker	2
Personnel access	Trip, slip hazards.	2	1. Clear any obstacles from around all work areas before and as work progresses. 2. Keep pathways clear of trip hazards.	Worker	3
Material's handling	Manual handling disorders.	2	1. Park delivery vehicle as close as safely possible to area of installation to minimize handling. 2. Physical fitness and warm up exercises will assist in avoiding manual handling injury. 3. Only carry what is a comfortable weight and do not overload yourself. Team lift if necessary. 4. Bend your knees when lifting not your back.	Worker	3
Operating plant	Plant failure	1	1. Plant to be properly maintained and checked before use. 2. Operator to wear seat belt. 3. Ensure roll over protection structure is in good working condition. 4. Ensure safety pins are fitted into quick hitch. 5. When lifting the correct certified lifting gear must be used. 6. Regularly maintain plant and machinery to manufacturer's requirements. Plant maintenance and logbook details to be kept current.	Worker	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Set out work area and check excavation levels.	Struck by road or construction plant.	1	1. Barricade area off or sign appropriately to restrict access and minimise plant working area. 2. All workers to wear hi-vis clothing or safety vests. 3. Operational flashing lights and reversing beepers.	Worker	2
	Striking underground services - electrocution	1	1. Before digging starts, make sure you know the exact location of any underground electrical cables, gas lines, water, sewerage and telecommunications cables. 2. Do not rely solely on site plans and drawings as these are sometimes not accurate or complete. 3. Seek assistance from the local services and distribution companies. 4. Use "Dial Before U Dig" -1100	Worker	2
	Persons falling, slipping or tripping over on site.	2	1. Maintain good house keeping. 2. Barricade off unsafe areas. 3. Clearly mark protruding objects using paint, caps or hazard tape. 4. Workers regularly clean boots to prevent slipping.	Worker	3
	Dust entering eyes	2	5. Suppress dust using water truck. 6. Eye protection to be worn when required.	Worker	3
Fuelling excavator	Fuel splashing into eyes. Fuel spill	2	1. Wear eye protection – eye wash in first aid kit. 2. Fuelling of excavator carried out off site.	Worker	3
Excavator operation	Engine failure	2	1. Warm up engine and hydraulic oil before starting work.	Worker	3
Excavating below 1.5m	Trench collapse	1	1. Ensure paperwork has been completed Notice to commence operations in trenches. WorkCover is to be notified 3 days prior to digging a trench over 1.5m deep. 2. Ensure trench is inspected by suitably qualified person if over 1.5m in depth. 3. Ensure trench is protected by appropriate trench protection.	Worker	2
Shoring of trench and inserting trench shields	Trench Collapse Crushing injury to anybody in the trench or even death.	1	1. Never allow workers to enter a trench or shaft which is greater than 1.5 metres deep unless it has been safely battered back, or it has been properly shored, or the workers are fully protected within a trench shield. 2. Do not allow machinery close to the edge of an excavation. If this is unavoidable because of buildings or other obstructions, use either a stronger ground support system or a method of excavation other than	Worker	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
			trenching. - A worker should never work alone in a trench unless there is another person close by who can render aid if necessary. - During routine inspections, the person in charge should be alert to certain unsafe conditions which are common during excavating, pipelaying and backfilling. - Surveillance of trench walls and support systems should be carried out frequently to ensure that: - Trench sides are not being undercut by the excavator bucket. - Supports are not being overstressed. - The ground does not begin to collapse into the trench. - Tension cracks do not appear along the trench top. - The trench walls do not sag under the increased pressure of the excavator. - To eliminate as much risk as possible always backfill as soon as possible to at least 1.5m. - Do not leave trenches open and not backfilled over night or during days of no work. If they have to be left open then barricade them accordingly. - Operators shall not enter a deep excavation until the excavation supervisor has assessed the excavation as safe to enter. - Industrial strength ladders should be used to enter a trench and ladders must be less than 30m apart. - Installation and removal of shoring is to take place outside of the excavation.		
Lifting shields into trench	Attaching hooks to the shield when on the ground, falling from shield or ladder used to get to the attachment points	2	- Place all shields on their side so as to not need a ladder to reach the attachment points. If using a ladder, set up ladder safely and check stability first. - Where possible allow the shield to be above the ground to act as a fence or barricade. (Minimum of 300mm as is required in the Code of Practice – Safety in Trenching Operations, but up to 900mm is a better barrier). - When approaching the shield to attach hooks, to enter the trench or to pour do either or both of the following;	Worker	3

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
			(a) Backfill between the shield and trench wall to at least 1.0m below ground level. (b) Use a ramp walkway with handrail and kick board, from ground level to the shield or manhole. Securely place the ramp on level ground and the edge of the shield.		
Entering trenches Using ladders	Fall from heights	1	1. Enter and leave trenches deeper than 1.5m by two means only; (a) Ladder outside the shield where the trench is backfilled to less than 1.5m (b) Ladder inside the shield with safe access of the type specified above 2. Make sure the ladder is commercial quality and is in good condition. Return the ladder to the Foreman, Supervisor or Manager if it is damaged or not suitable for the task. 3. The ladder must be tall enough to extend at least 900mm above the level needed to stand or step off. 4. The ladder must be securely positioned at the bottom, eg push it into bedding or backfilling, or tie it to the shield. 5. Restrain the top so that it cannot slide away, eg tied to the top of the shield. 6. Make sure the ladder is at least 900mm above the work level or where you step off. 7. Use the top 900 mm of the ladder to hold onto; do not stand on the top 900 mm of the ladder. 8. If the trench is deeper than 1.5m position the ladder within the shield with a shield cover always. 9. If the trench is less than 1.5m deep the ladder can be outside the shield but preferably against the shield.	Worker	2
Delivery of bedding & crushed rock	Vehicle movements	1	1. Workers wear hi-vis clothing or safety vests. 2. Correct signing and barricading in place.	Worker	2
Placement of bedding	Falling into trench	2	1. Provide ladder access in and out of excavations/trench shields. 2. Barricade trench when not in use.	Worker	3
	Fumes	2	1. Ensure dimension of trench is adequate for ventilation of diesel equipment.	Worker	3

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
	Personnel struck by back fill	1	1. Use spotter during placement 2. Wear hard hats and safety glasses.	Worker	2
	Back or muscle injury	2	1. Correct manual handling techniques if using shovels.	Worker	3
	Uneven ground	3	1. Plant operator to be aware of uneven surfaces and if possible ensure level ground prior to plant moving bedding and crushed rock.	Worker	3
Unloading pipes	Road Traffic	1	1. Workers wear hi-vis clothing or safety vests. 2. Correct signing and barricading in place.	Worker	2
	Pipes slipping off truck	2	1. Pipes are securely tied onto tray of truck.	Worker	3
	Moving pipes off truck	2	1. Use correct manual handling techniques. 2. Use a mechanical lifting device if it is available.	Worker	3
Attaching, Lifting and Placing Loads near and in a Deep Trench. Trench shield and pipes	Trench collapse Injury to workers	1	1. The operator and worker directing the movement of a shield must work together and carry out the activity safely. The worker directing the load must stand well clear of the trench and the moving load. 2. Do not stand or balance on the shield, under any circumstances. 3. The unprotected trench where pipe are being laid is the most dangerous. Only competent pipe layers or helpers allowed in this area. 4. The area should be kept to a minimum. 5. The edge of the trench for a width of at least 2m should be kept smooth and clear of tools and pipes that can be tripped over. - The pipe laying activity should be planned safely. - Consider the method of lifting and lowering the pipes, to eliminate as much as possible getting close to the edge of the trench. - Guide pipes with a rope standing away from the edge. - Wherever possible lower the pipe through the shield, this allows some protection by the shield if it is above the ground. - Ensure that no one is under the load as it is being lowered into the trench.	Worker	2
Laying Pipes	Falling into trench	2	1. Provide ladder access in and out of excavations. 2. Barricade trench when not in use.	Worker	3
Manual lift	Lifting into trench – muscle injury	2	1. Use correct manual handling techniques. 2. Use a mechanical lifting device if it is available.	Worker	3
Machine lift	Crushing of plant or personnel by pipes.	1	1. Ensure competent dogman secures load. 2. Use certified lifting chains and shackles. 3. Check lifting capacity ad SWL prior to lift.	Worker	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
			4. Remove excavation bucket prior to lifting pipes. 5. Do not slew load over personnel. 6. Competent dogman required whilst lifting.		
	Striking overhead services-electrocution	1	Operators need to keep; 1. At least 3 metres from 132,000 volt lines. 2. At least 6 metres from 132,000 to 330,000volt lines. 3. Areas closer than 3m are no go zone. 4. Use certified spotter if working near powerlines. 5. (3m to 6.4m). 6. Clarify safe working distances with site manager. 7. If in doubt of line voltage keep at least 8 metres away.	Worker	2
Clean Up	Various for current and following Worker	2	1. Leave the site in a clean and safe manner Note; Be aware that a back-charge may apply if this is not done.	Worker	3

ENVIRONMENTAL RISK ASSESSMENT & CONTROL MEASURES THAT MAY BE REQUIRED ON SITE DEPENDING ON THE ACTIVITY					
Work Activity Risk	Hazard	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class after Controls in place
Emergencies	Accident/ Incident Fire Spill	1	1. Personnel have been advised at toolbox meeting and are prepared in the event of an incident occurring. 2. Emergency numbers available on site. 3. Fire Extinguishers/First Aid Kits/Spill Kits on site.	Supervisor All	2
Air Quality	Exhaust Fumes Dust	2	1. Control exhaust emissions eg. Regular maintenance, replace old machinery, fit appropriate emission control measures. 2. Dampen milled surface with water cart. 3. Use water cart where required.	Supervisor All	3
Bitumen/Prime	During heavy rain, recently placed prime or bitumen could emulsify and wash into surrounding waterways.	1	1. Plan prime or bitumen operation so that it is conducted during fine weather. 2. Have material available to clean up a spill if it occurs and dispose of contaminated material accordingly. 3. Protect storm drains during and after application to prevent pollution - have spill clean-up materials available.	Supervisor All	2
Community Relations	Inconvenience as a result of works/traffic management	2	1. Identify from the list of potentially impacted parties who may be impacted by the work being undertaken. 2. Letter drops could be considered (unless specifically required) to notify residents/businesses etc. about works to be undertaken and if it has an effect on them.	Supervisor All	3
Flora & Fauna	Some works may require the removal of vegetation. Native fauna may be at risk from excavation works	2	1. Assess work area for significant vegetation: size of trees, faunal habitat - define work and exclusion areas using fencing and signage. 2. Machinery is to be thoroughly washed down prior to leaving site where noxious weeds are present.	Supervisor All	3
Heritage & Archaeology	Unmonitored excavation at significant sites can lead to loss or destruction of valuable artifacts/relics, and disturbance of historical sites. including cultural heritage and/or archaeological significance	1	1. Undertake a survey of the site to identify any areas of significance. The client's representative may undertake this. Government bodies may be able to assist. 2. Develop a project specific procedure based on the findings and recommendations. 3. Installation of exclusion fencing and signage if required.	Supervisor All	2

Work Activity Risk	Hazard	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class after Controls in place
Noise Pollution, Community Relations & Vibration	Sensitivity to noise - to public & fauna	1	1. Work undertaken during “normal” hours where possible. 2. Affected residents notified in person or in writing. 3. Plant and equipment regularly serviced & fitted with efficient mufflers. 4. Magnitude of vibration reduced by using smaller plant etc.	Supervisor All	2
Soil Management & Water Quality	Soil needs to be managed to ensure that; there are no off-site impacts. Damage to stockpile areas (not prepared properly). Mud deposited onto roadways.	2	1. Wash down machinery in only nominated areas. 2. Not within 20 metres of water courses. 3. Use only approved and nominated stockpile areas. 4. Build bund wall out of material available on site. 5. Drivers to check vehicle tyres/bodies for build up of mud when leaving stockpile sites and remove if necessary.	Supervisor All	3
Fuels and Chemicals	Contamination of waterways due to spillages of oil, degreasers, diesel, etc.	1	1. Store fuel and chemicals in accordance with relevant standards/guidelines. 2. Minimise storage on site and locate storage facilities away from drains and waterways. 3. All containers clearly labelled and Safety Data Sheets are available on site. 4. All drums stored securely and in a bunded area. 5. Spill kits or alternative spill containment materials available on site. 6. Regular maintenance of plant hoses. 7. Use correct refuelling equipment.	Supervisor All	2
Visual Impact Litter and Amenities Waste Management	Litter such as paper, plastic, and other refuse as well as being a possible safety risk may be carried off-site or washed into waterways resulting in pollution and danger to native fauna. Litter is also visually obtrusive. Production generated lighting, litter, mud, dirt and dust may also impact residents.	2	1. Use site induction to communicate the requirement for a tidy site. Also stress consideration for local residents. 2. If as a result of our works, dirt, mud, or litter affects a resident, then we will rectify this problem immediately. 3. Inspect the site frequently and insist on tidiness. Ensure public roads are free of dirt/mud from the work site. 4. Broken-out paved surfaces shall be repaired as soon as possible. 5. Remove all rubbish from site each day. 6. Waste asphalt will be recycled where possible.	Supervisor All	3

Personal Protective Equipment

All site personnel must wear the required PPE specified in the controls above, such as:

Safety Hi-vis Vests/Shirts	Safety Boots	Sunscreen	Broad Brimmed Sun Hat	Sunglasses	Gloves specific to the task	White reflective overalls/night
Hard Hat where required	Safety Glasses	Hearing protection where required	Long clothing	Dust masks	Respiratory equipment if required	

Workers may require Task Specific Training depending on activity each worker will be required to fulfill.

All workers shall hold a current Construction Induction Card						
- If required Client Site Induction - Activity Induction (SWMS & SWP's)	Vehicle Licences (Car or Truck depending on class of vehicle or mobile plant driven)	Plant Operators licences or competency assessments where required	Competencies for Work Zone Traffic Management			
First Aid certificate training	Manual Handling training	Fire Extinguisher training	Fatigue Management Training			

A Site Specific Induction & Toolbox Meeting will be held each day prior to the start of each job.

Legislation & Codes of Practice (refer to company Legislation register for applicable Australian Standards applicable to works)

Work Health & Safety Act 2012 and Work Health & Safety Regulations 2022

Codes of Practice

Confined Spaces - 2020	Hazardous Manual Tasks - 2020	Managing Noise & Preventing Hearing Loss at Work-2020	Managing Risks of Plant in the Workplace - 2018
Construction Work - 2020	How to Manage WH&S Risks - 2018	Managing Risk of Falls at Workplaces - 2020	Managing Electrical Risks in the Workplace -2019
Excavation Work - 2020	First Aid in the Workplace - 2018	Managing Work Environment & Facilities - 2020	WHS Consultation Co-operation & Co-ordination - 2018
Welding Processes - 2021	How to Safely Remove Asbestos - 2020	Managing Risk of Hazardous Chemicals - 2018	Safe Design of Structures - 2018

Plant & Equipment for this activity

- All plant is inspected prior to sending out to site
- Daily maintenance & service checks.
- Daily pre-start checks are conducted on all plant and trucks by the operators and daily checklist filled out.
- Regular servicing at 250hr intervals or as per manufactures specifications. Service History available on request to the office.

List of plant for this Activity

Engineering details/certificates/regulatory approvals/permits/electrical isolation (if applicable):

Implementation, Monitoring and Review of SWMS

The Supervisor on site has the responsibility to:

- Brief each team member on this SWMS before commencing work.
- Ensure control measures have been implemented prior to the start of works.
- Observe work being carried out.
- If controls documented in SWMS are not adequate, stop the work, review the SWMS, adjust as required and re-brief the team.
- Ensure team knows that work is to immediately stop if the SWMS is not being followed.
- Retain this SWMS on site for the duration of the high-risk construction work.
- All persons involved in these particular tasks on site must sign this document as read and understood.

SWMS attendance sheet — I have been given the opportunity to provide input into the development and review of this SWMS

Name (please print clearly)	Signature	Company	Date

Daily pre-start meetings will be held to ensure all workers are informed of control measures and additional safety hazards & controls to be noted on toolbox form.